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Department of Electronics & Communication Engineering | GATE Preparation Library

Comprehensive GATE Notes: Signals and Systems

Paper Code: EC • Section 2: Continuous & Discrete Signals and Systems (8-11 Marks)

Subject Code: EC-SIG • **Target Exam:** GATE ECE • **Weightage:** 8-11 Marks

1. LTI System Properties & Convolution

- **LTI System Fundamentals:** A linear time-invariant system is completely characterized by its impulse response h(t) [or h[n]]. Output y(t) = x(t) * h(t) = int_{-inf}^{inf} x(tau) h(t - tau) dtau.
- **Causality:** Continuous system is causal iff h(t) = 0 for t < 0. Discrete system is causal iff h[n] = 0 for n < 0.
- **BIBO Stability:** Continuous system is BIBO stable iff int_{-inf}^{inf} |h(t)| dt < inf. Discrete system is BIBO stable iff sum_{n=-inf}^{inf} |h[n]| < inf.
- **Convolution of Rectangular Pulses:** The convolution of two identical rectangular pulses of width T and height A is a symmetric triangular pulse of base width 2T, centered at the sum of their centers, with peak height A^2 * T.

Topic / Law	Key Formula & Equation	Variables / Notation	GATE Application / Traps
Continuous Convolution	**y(t) = int_{-inf}^{inf} x(tau) h(t - tau) dtau**	tau = Dummy Integration Variable	Commutative, distributive, and associative operator
Discrete Convolution	**y[n] = sum_{k=-inf}^{inf} x[k] h[n - k]**	k = Dummy Summation Variable	Length of result = N1 + N2 - 1 for finite sequences
Continuous BIBO Stability	**int_{-inf}^{inf} |h(t)| dt < inf**	h(t) = Impulse Response	Equivalent to all transfer function poles strictly in LHP

GATE PYQ (2 Marks) - Convolution of Unequal Rectangular Pulses
Question: Let x(t) = u(t) - u(t-2) and h(t) = u(t) - u(t-3). Find the maximum value of y(t) = x(t) * h(t) and the time interval at which it occurs.

Step 1: x(t) is width T1 = 2, height A = 1. h(t) is width T2 = 3, height B = 1.
Step 2: The convolution of two unequal rectangles is a trapezoid starting at t = 0 + 0 = 0.
Step 3: Rises linearly until t = min(T1, T2) = 2. Plateau from t = 2 to t = max(T1, T2) = 3. Falls to zero at t = 5.
Step 4: Plateau peak = A * B * T1 = 1 * 1 * 2 = **2.0** over **2 <= t <= 3**.

2. Laplace & Z-Transforms: ROC & Transfer Functions

- **Region of Convergence (ROC) Rules:**
- ROC cannot contain any poles.
- For right-sided signals (causal): ROC is Re{s} > sigma_max (Laplace) or |z| > r_max (Z-transform).
- System is BIBO stable iff the imaginary j-omega axis lies inside Laplace ROC, or the unit circle |z| = 1 lies inside Z-transform ROC.

Topic / Law	Key Formula & Equation	Variables / Notation	GATE Application / Traps
Laplace IVT / FVT	**x(0+) = lim_{s->inf} sX(s) | x(inf) = lim_{s->0} sX(s)**	s = Complex Frequency	FVT only valid if system steady-state exists
Z-Transform FVT	**x[inf] = lim_{z->1} (z - 1) X(z)**	z = Complex Variable	All poles of (z-1)X(z) must strictly reside inside unit circle
Causal + Stable Z condition	**All poles strictly inside |z| < 1**	|z| = Unit Circle	Encloses unit circle with outward radial ROC

GATE PYQ (1 Mark) - Stability and Causality from ROC
Question: A causal LTI discrete system has H(z) = (z - 0.5) / [(z - 0.8)(z + 1.2)]. Is the system BIBO stable?

Step 1: Poles are z_1 = 0.8 and z_2 = -1.2.
Step 2: Since system is CAUSAL, ROC must be exterior to outermost pole: ROC is |z| > 1.2.
Step 3: For BIBO stability, ROC must include the unit circle |z| = 1.
Step 4: Since |z| > 1.2 does not include |z| = 1, the causal system is **UNSTABLE**.

Essential GATE Exam Pitfalls & Scoring Traps

- **Important:** Before applying the Final Value Theorem, ALWAYS check that no poles lie in the RHP or on the j-omega axis.
- **Important:** In sampling theorem, minimum sampling rate is Nyquist rate f_s = 2 * f_max; Nyquist interval is T_s = 1 / (2 * f_max).